

Krishna Chaitanya.V

Data Engineer

krishna@workwebmail.com | (309) 612-8656 | USA | [LinkedIn](#)

Summary

Results-driven Data Engineer with 3+ years of experience in big data processing, ETL development, and analytics. Proficient in Apache Spark, AWS Glue, Redshift, and Azure Synapse, optimizing large-scale data pipelines and enhancing real-time analytics. Skilled in Python, SQL, Power BI, and Tableau, delivering data-driven insights. Experienced in healthcare and financial data processing, ensuring GDPR compliance, data governance, and anomaly detection to improve operational efficiency and decision-making.

Technical Skills

- Big Data & Data Engineering:** Hadoop HDFS, Apache Spark (Databricks, AWS EMR), AWS Glue, Step Functions, AWS Redshift
- Cloud Platforms:** AWS (Glue, Redshift, Lake Formation, EMR, QuickSight, IAM), Azure (Data Factory, Synapse Analytics)
- Data Analysis & Visualization:** Python (Pandas, NumPy, PySpark), SQL, Tableau, Amazon QuickSight, Power BI, Advanced Excel (Pivot Tables, Regression Analysis), Statistical Analysis (t-tests, ANOVA)
- ETL & Data Integration:** AWS Glue, Step Functions, Azure Data Factory, Azure Synapse Analytics, Apache Airflow
- Data Governance & Security:** AWS Lake Formation, IAM Policies, GDPR Compliance
- Machine Learning & AI:** Anomaly Detection (Isolation Forests, DBSCAN), NLP, Clustering (K-Means), Random Forest, Decision Trees
- Programming & Scripting:** Python, SQL
- Workflow Optimization:** Schema Evolution, Partitioning Strategies, Data Caching
- Reporting & Dashboards:** Tableau, Power BI, Looker, Amazon QuickSight

Professional Experience

Data Engineer, HSBC

06/2023 – Present | Remote, USA

- Designed a real-time stock market data pipeline using Apache Kafka and AWS Kinesis, processing 5 million transactions per second to enable high-frequency analytics, anomaly detection, and regulatory compliance.
- Developed scalable ETL workflows with Apache Spark and AWS Glue, optimizing data transformation for 500 billion+ records in AWS Redshift, reducing query latency by 60% through partitioning, compression, and optimized columnar storage.
- Built Python-based data validation scripts with Pandas and PySpark, detecting 95% of anomalies in stock market feeds before ingestion into AWS S3, ensuring data accuracy, consistency, and compliance with financial governance policies.
- Applied machine learning algorithms like isolation forests and DBSCAN for anomaly detection, reducing false positives in fraud detection by 30% and improving the reliability of financial transaction data.
- Automated data workflows using Apache Airflow, streamlining ingestion, transformation, and anomaly detection across 100+ sources to enhance data availability, SLA compliance, and operational efficiency.
- Engineered a scalable data lake architecture with AWS S3 and AWS Glue, improving secure storage, metadata management, and retrieval of structured and unstructured financial data while optimizing query response times.
- Optimized complex SQL queries in Amazon Redshift with window functions and indexing, improving analytical query performance by 40% for real-time stock market trend monitoring in Looker dashboards.

Data Analyst, Primera Medical Technologies

01/2021 – 07/2022 | Remote, India

- Conducted healthcare cost trend analysis using Python (Pandas, NumPy), improving cost prediction accuracy by 12% by analyzing patient visits, treatments, and diagnostic test patterns to identify cost drivers and optimize financial planning.
- Utilized Advanced Excel (pivot tables, regression analysis) to compute utilization rates and treatment costs, enhancing forecasting accuracy by 20% and enabling better resource allocation for healthcare services based on historical cost patterns.
- Integrated healthcare data from EHR and billing systems using Azure Data Factory and Azure Synapse Analytics, accelerating data ingestion and query performance by 30% for large-scale cost analysis and improving data integration workflows.
- Applied machine learning algorithms (Random Forest, Decision Trees) for data cleansing and anomaly detection, reducing inconsistencies by 22%, improving data reliability, and ensuring high-quality datasets for advanced healthcare analytics.
- Applied clustering techniques (K-Means) to segment patient cost patterns, identifying high-risk groups and enabling targeted cost-reduction strategies that optimized medical resource distribution and improved financial efficiency.
- Utilized NLP techniques to categorize unstructured medical records, increasing data accessibility by 30% and streamlining healthcare data processing for improved insights into patient histories, diagnoses, and treatment costs.
- Designed interactive Tableau and Power BI dashboards to visualize cost patterns, utilization trends, and cost-saving opportunities, leading to a 25% increase in stakeholder engagement and a 15% reduction in healthcare expenditures.

Education

Master of Science, University of the Cumberlands
Information Systems

08/2022 – 05/2024 | Williamsburg, KY, USA